

LAB 5 – QUERIES AND SUBQUERIES

1. SQL Query

A **query** is a SQL statement used to retrieve or manipulate data from a database.

Basic Syntax

```
SELECT column1, column2
```

```
FROM table_name
```

```
WHERE condition;
```

Example

```
SELECT Name, Marks
```

```
FROM Student
```

```
WHERE Marks > 80;
```

This displays the name and marks of students who scored more than 80.

5. Subquery

A **subquery** is a query written inside another SQL query.

It is also called a **nested query** or **inner query**.

General Syntax

```
SELECT column
```

```
FROM table
```

```
WHERE column operator
```

```
(
```

```
    SELECT column
```

```
    FROM table
```

```
    WHERE condition
```

```
);
```

A subquery generally consists of:

- **Inner query** – executes first and produces a result.

- **Outer query** – uses the result of the inner query.

Example

SELECT Name, Marks

FROM Student

WHERE Marks > (

SELECT AVG(Marks)

FROM Student

);

The inner query calculates the average marks. The outer query then displays students whose marks are greater than that average.

```
mysql> CREATE TABLE Student (
-> Student_ID INT,
-> Name VARCHAR(50),
-> Department VARCHAR(10),
-> Marks INT,
-> City VARCHAR(50)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Student (Student_ID, Name, Department, Marks, City) VALUES
-> (101, 'Rahul', 'CSE', 85, 'Delhi'),
-> (102, 'Priya', 'IT', 92, 'Mumbai'),
-> (103, 'Amit', 'CSE', 76, 'Delhi'),
-> (104, 'Neha', 'ECE', 88, 'Pune'),
-> (105, 'Riya', 'IT', 95, 'Delhi'),
-> (106, 'Karan', 'CSE', 68, 'Mumbai'),
-> (107, 'Anjali', 'ECE', 81, 'Pune');
Query OK, 7 rows affected (0.01 sec)
Records: 7 Duplicates: 0 Warnings: 0
```

PART A – 7 SQL QUERIES

Query 1: Display all students

SELECT *

FROM Student;

Purpose: Displays all columns and all records from the Student table.

```
mysql> SELECT *
-> FROM Student;
+-----+-----+-----+-----+-----+
| Student_ID | Name | Department | Marks | City |
+-----+-----+-----+-----+-----+
| 101 | Rahul | CSE | 85 | Delhi |
| 102 | Priya | IT | 92 | Mumbai |
| 103 | Amit | CSE | 76 | Delhi |
| 104 | Neha | ECE | 88 | Pune |
| 105 | Riya | IT | 95 | Delhi |
| 106 | Karan | CSE | 68 | Mumbai |
| 107 | Anjali | ECE | 81 | Pune |
+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)
```

Query 2: Display selected student details

SELECT Name, Department, Marks

FROM Student;

Purpose: Displays only the student's name, department and marks.

```
mysql> SELECT Name, Department, Marks
-> FROM Student;
+-----+-----+-----+
| Name | Department | Marks |
+-----+-----+-----+
| Rahul | CSE       | 85    |
| Priya | IT        | 92    |
| Amit  | CSE       | 76    |
| Neha  | ECE       | 88    |
| Riya  | IT        | 95    |
| Karan | CSE       | 68    |
| Anjali | ECE      | 81    |
+-----+-----+-----+
7 rows in set (0.00 sec)
```

Query 3: Display students scoring more than 80

SELECT Name, Marks

FROM Student

WHERE Marks > 80;

Purpose: Displays students who scored more than 80 marks.

```
mysql> SELECT Name, Marks
-> FROM Student
-> WHERE Marks > 80;
+-----+-----+
| Name | Marks |
+-----+-----+
| Rahul | 85    |
| Priya | 92    |
| Neha  | 88    |
| Riya  | 95    |
| Anjali | 81    |
+-----+-----+
```

Query 4: Display CSE students

SELECT Name, Marks

FROM Student

WHERE Department = 'CSE';

Purpose: Displays students belonging to the CSE department.

```
mysql> SELECT Name, Marks
-> FROM Student
-> WHERE Department = 'CSE';
+-----+-----+
| Name | Marks |
+-----+-----+
| Rahul | 85    |
| Amit  | 76    |
| Karan | 68    |
+-----+-----+
3 rows in set (0.00 sec)
```

Query 5: Display students from Delhi scoring more than 80

SELECT Name, Marks

FROM Student

WHERE City = 'Delhi'

AND Marks > 80;

Purpose: Displays students who are from Delhi and scored more than 80.

```
mysql> SELECT Name, Marks
-> FROM Student
-> WHERE City = 'Delhi'
-> AND Marks > 80;
+-----+-----+
| Name | Marks |
+-----+-----+
| Rahul | 85    |
| Riya  | 95    |
+-----+-----+
2 rows in set (0.00 sec)
```

Query 6: Find the highest marks

SELECT MAX(Marks) AS Highest_Marks

FROM Student;

Purpose: Finds the highest marks obtained by a student.

```
mysql> SELECT MAX(Marks) AS Highest_Marks
-> FROM Student;
+-----+
| Highest_Marks |
+-----+
| 95            |
+-----+
1 row in set (0.01 sec)
```

Query 7: Find the average marks

SELECT AVG(Marks) AS Average_Marks

FROM Student;

Purpose: Calculates the average marks of all students.

```
mysql> SELECT AVG(Marks) AS Average_Marks
-> FROM Student;
+-----+
| Average_Marks |
+-----+
| 83.5714       |
+-----+
1 row in set (0.00 sec)
```

PART B – 7 SUBQUERIES

Subquery 1: Students scoring above average

```
SELECT Name, Marks
FROM Student
WHERE Marks > (
    SELECT AVG(Marks)
    FROM Student
);
```

Purpose: Displays students whose marks are greater than the overall average marks.

Working:

Inner query:

```
SELECT AVG(Marks)
FROM Student;
```

Finds the average marks.

Outer query:

```
SELECT Name, Marks
FROM Student
WHERE Marks > average;
```

Displays students scoring above the average.

```
mysql> SELECT Name, Marks
-> FROM Student
-> WHERE Marks > (
-> SELECT AVG(Marks)
-> FROM Student
-> );
```

Name	Marks
Rahul	85
Priya	92
Ileha	88
Riya	95

```
4 rows in set (0.01 sec)
```

Subquery 2: Student with the highest marks

```
SELECT Name, Marks
FROM Student
WHERE Marks = (
    SELECT MAX(Marks)
    FROM Student
```

);

Purpose: Finds the student or students who obtained the highest marks.

```
mysql> SELECT Name, Marks
-> FROM Student
-> WHERE Marks = (
->   SELECT MAX(Marks)
->   FROM Student
-> );
```

Name	Marks
Riya	95

1 row in set (0.00 sec)

Subquery 3: Students scoring more than Rahul

SELECT Name, Marks

FROM Student

WHERE Marks > (

SELECT Marks

FROM Student

WHERE Name = 'Rahul'

);

Purpose: Displays students whose marks are greater than Rahul's marks.

Working:

The inner query finds Rahul's marks. The outer query compares every student's marks with Rahul's marks.

```
mysql> SELECT Name, Marks
-> FROM Student
-> WHERE Marks > (
->   SELECT Marks
->   FROM Student
->   WHERE Name = 'Rahul'
-> );
```

Name	Marks
Priya	92
Neha	88
Riya	95

Subquery 4: Students from the same department as Rahul

SELECT Name, Department

FROM Student

WHERE Department = (

SELECT Department

FROM Student

WHERE Name = 'Rahul'

);

Purpose: Displays students who belong to the same department as Rahul.

```
mysql> SELECT Name, Department
-> FROM Student
-> WHERE Department = (
->   SELECT Department
->   FROM Student
->   WHERE Name = 'Rahul'
-> );
```

Name	Department
Rahul	CSE
Amit	CSE
Karan	CSE

3 rows in set (0.00 sec)

Subquery 5: CSE students scoring above the CSE average

SELECT Name, Marks

FROM Student

WHERE Department = 'CSE'

AND Marks > (

SELECT AVG(Marks)

FROM Student

WHERE Department = 'CSE'

);

Purpose: Finds CSE students whose marks are greater than the average marks of CSE students.

```
mysql> SELECT Name, Marks
-> FROM Student
-> WHERE Department = 'CSE'
-> AND Marks > (
->   SELECT AVG(Marks)
->   FROM Student
->   WHERE Department = 'CSE'
-> );
```

Name	Marks
Rahul	85

1 row in set (0.00 sec)

Subquery 6: Students whose department is among Mumbai students

SELECT Name, Department

FROM Student

WHERE Department IN (

SELECT Department

FROM Student

WHERE City = 'Mumbai'

);

Purpose: First finds the departments of students from Mumbai and then displays students belonging to those departments.

Important:

IN is used when the subquery can return **multiple values**.

```
mysql> SELECT Name, Department
-> FROM Student
-> WHERE Department IN (
-> SELECT Department
-> FROM Student
-> WHERE City = 'Mumbai'
-> );
```

Name	Department
Rahul	CSE
Priya	IT
Amit	CSE
Riya	IT
Karan	CSE

5 rows in set (0.00 sec)

Subquery 7: Students scoring higher than all ECE students

SELECT Name, Marks

FROM Student

WHERE Marks > ALL (

SELECT Marks

FROM Student

WHERE Department = 'ECE'

);

Purpose: Displays students whose marks are greater than every ECE student's marks.

Important:

ALL means the condition must be true for **every value returned by the subquery**.

```
mysql> SELECT Name, Marks
-> FROM Student
-> WHERE Marks > ALL (
-> SELECT Marks
-> FROM Student
-> WHERE Department = 'ECE'
-> );
```

Name	Marks
Priya	92
Riya	95

2 rows in set (0.01 sec)

6. IN and ALL in Subqueries

IN

IN is used to compare a value with multiple values returned by a subquery.

```
WHERE Department IN (  
    SELECT Department  
    FROM Student  
    WHERE City = 'Mumbai'  
);
```

ALL

ALL compares a value with every value returned by a subquery.

```
WHERE Marks > ALL (  
    SELECT Marks  
    FROM Student  
    WHERE Department = 'ECE'  
);
```

Easy Difference

Operator Meaning

=	Equal to one value
IN	Matches any value in a list
ALL	Condition must satisfy every returned value

Quick Revision

7 SQL Queries

1. SELECT *
2. SELECT specific columns
3. WHERE Marks > 80
4. WHERE Department = 'CSE'

5. WHERE City = 'Delhi' AND Marks > 80
6. MAX(Marks)
7. AVG(Marks)

7 Subqueries

1. Marks greater than average
2. Student with maximum marks
3. Marks greater than Rahul
4. Same department as Rahul
5. Above CSE average
6. Department IN departments of Mumbai students
7. Marks greater than ALL ECE students

Remember the basic structure:

